Deep Reinforcement Learning

Planning & Optimal Control

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Content

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Where are we?

Chapter	Topic	Content
	Basics & tabular methods	
1	Multi-armed bandits	Exploration-exploitation dilemma
2	Markov decision processes	Dynamics, rewards, policies
3	Dynamic programming	Optimal decision making with *full knowledge*
4	Monte Carlo methods	*Data-driven learning* from entire episodes
5	Temporal difference learning & Q-learning	*Data-driven learning* from individual transitions
	Deep-learning-based methods	
6	**Brief introduction to deep learning**	**The basics for what comes next**
7	Deep Q-learning	
8	Policy gradients	
9	Actor-critic algorithms	
10	Advanced algorithms	
	Model-Based Control	
11	Planning & optimal control	

Chapter	Topic	Content
12	Model-based reinforcement learning	
13	Exploration	
	Advanced Topics	
14	Offline reinforcement learning	
15	Transfer learning	
16	Multi-agent reinforcement learning	
17	Behavior cloning	

Lecture contents

References